R vs East & nQuery

Utilisation de R comme alternative aux logiciels propriétaires East et nQuery pour le calcul de Sample Size.

AROCENA Thomas

vendredi 16 janvier 2026

Résumé Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sagittis posuere ligula sit amet lacinia. Duis dignissim pellentesque magna, rhoncus congue sapien finibus mollis. Ut eu sem laoreet, vehicula ipsum in, convallis erat. Vestibulum magna sem, blandit pulvinar augue sit amet, auctor malesuada sapien. Nullam faucibus leo eget eros hendrerit, non laoreet ipsum lacinia. Curabitur cursus diam elit, non tempus ante volutpat a. Quisque hendrerit blandit purus non fringilla. Integer sit amet elit viverra ante dapibus semper. Vestibulum viverra rutrum enim, at luctus enim posuere eu. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

Comparaison I : Sample size pour tester le hazard ratio dans un groupe à 2 bras.

Méthode de Schoenfeld

Le calcul de la taille d'échantillon de Schoenfeld, comme les autres méthodes de log rank se fait sous les hypothèses de loi exponentielles et de proportional hazard qui ne sont pas complètement satisfaisantes dans certain cas mais complètement ancré dans l'analyse de survie en générale.

Prenons α l'erreur de type I et β l'erreur de type II. Alors, en notant e le nombre d'evènement nécessaire, la formule (simple) de Schoenfeld donne :

$$e = 2 \times \frac{2\left(z_{1-\alpha/2} + z_{1-\beta}\right)^2}{\log(\mathrm{HR})} \tag{1}$$

Où z_x est le z-score de x, égale au quantile $(1-x)$ de la distribution normale standard.

Méthode de Lakatos

blabla

Paramètres

On va comparer les resultats de Rpact et de East sur l'ensemble de test suivant :

$\alpha \in \{0.01, 0.05, 0.1, 0.20, 0.49\}$

$\text{power} \in \{0.51, 0.8, 0.9, 0.99\}$

$\text{HR} \in \{0.1, 0.5, 0.7, 0.9, 0.99\}$

$\text{Survie à 3 ans} \in \{0.1, 0.3, 0.6, 0.9\}$

This is a block with gray background and slightly rounded corners.

Rpact

On utilise la fonction `getSampleSizeSurvival` pour estimer le nombre d'évènements et de patients nécessaires.

```
Attaching package: 'rlang'
```

```
The following objects are masked from 'package:purrr':

    %@%, flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```

East

```
New names:
Rows: 76 Columns: 24
── Column specification ────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
New names:
Rows: 96 Columns: 24
── Column specification ────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
New names:
Rows: 96 Columns: 24
── Column specification
```

```
────────────────────────────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
New names:
Rows: 100 Columns: 24
── Column specification
────────────────────────────────────────────────────────────────────────────────
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
• `` → `...24`
```

nQuery

Rashnu

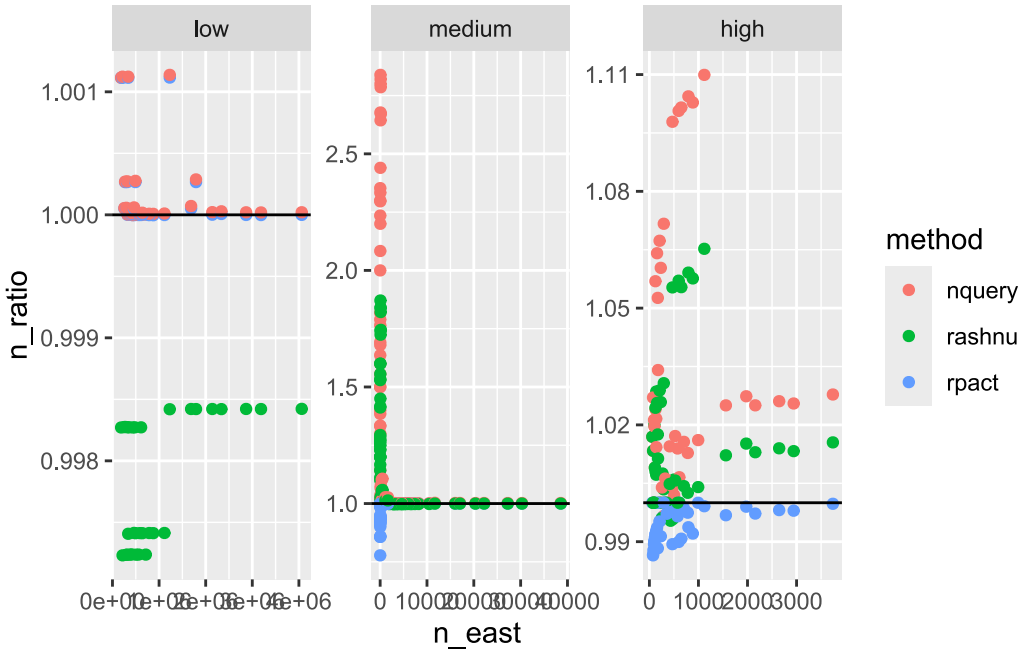
Inter-method comparison

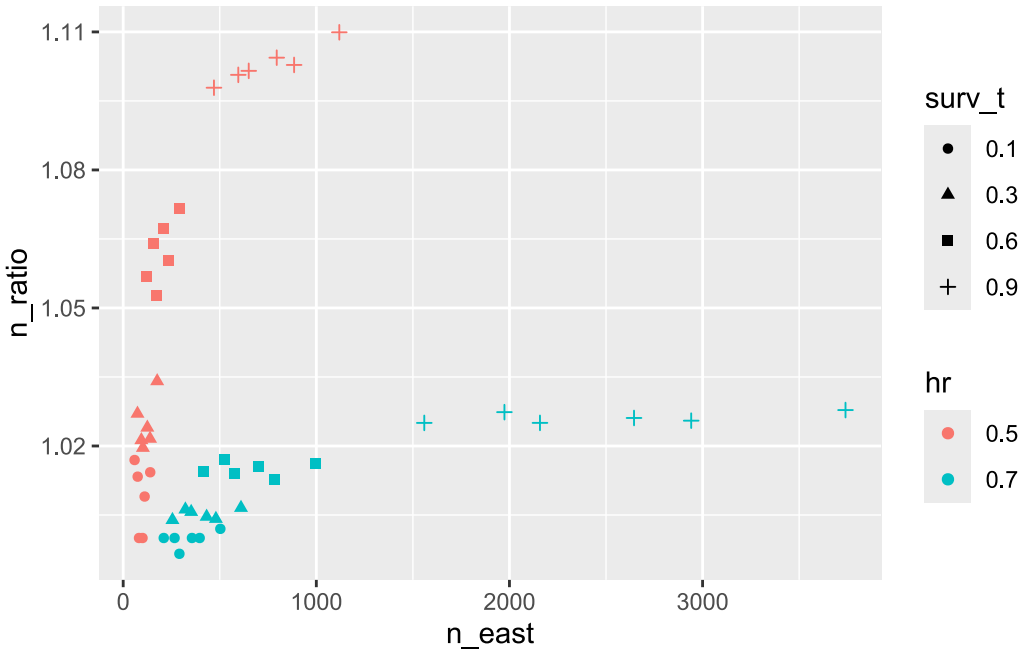
```

Attaching package: 'rlang'
```

```
The following objects are masked from 'package:purrr':

    %@%, flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```





Lakatos and Lan (1992)

rpact

East

```
New names:
Rows: 18 Columns: 24
── Column specification
────────────────────────────────────────────────────────────────────────────────────────────────────────────────
```

```
Delimiter: "," chr
(4): ID, Design Type, Test Type, Var (Log HR) dbl (19): Sr. No., No. of
Looks,
Specified α, Power, nt/nc, Sample Size, Exp ... lgl (1): ... 24
ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
Specify the column types or set `show_col_types = FALSE` to quiet this
message.
• `` → `...24`
```

nQuery

Rashnu

Comparision

			n_rpact	n_east	n_query	n_rashnu	n_L	n_Fn_RGS	power_rpact	power_L	power_Fn_RGS
0.8	0.667	1	1588	1593	1142	1620	1617	1628	1640	90.2	90.4	90.6
		5	1979	1984	1414	2022	2017	2024	2046	90.6	90.7	91.0
		9	2671	2679	1902	2728	2724	2709	2764	90.3	90.1	90.7
	0.500	1	601	604	356	638	638	649	664	90.1	90.6	91.2
		5	749	753	440	798	798	807	831	90.7	91.0	91.8
		9	1011	1017	590	1078	1079	1081	1124	90.1	90.2	91.2
	0.250	1	181	182	78	230	230	241	269	91.9	93.0	95.3
		5	225	227	94	290	289	299	338	92.2	93.0	95.5
		9	304	306	126	392	392	401	459	91.6	92.2	95.1
0.2	0.667	1	361	362	308	362	360	370	363	89.6	90.4	89.8
		5	414	415	342	416	414	419	418	90.5	90.8	90.8
		9	527	528	422	530	528	509	534	89.8	88.7	90.1
	0.500	1	133	134	102	134	134	144	138	89.7	91.7	90.5
		5	154	155	112	156	156	164	161	89.9	91.3	90.8
		9	196	197	138	200	200	200	207	89.7	89.7	90.7
	0.250	1	40	40	26	44	43	53	48	90.2	95.1	93.0
		5	46	47	28	52	51	61	58	90.6	94.8	93.8
		9	59	60	32	68	66	74	76	90.2	93.1	93.7

Comparison II : Group sequential design

Rpact

see rpact doc

East

J'ai observé une variabilité entre 2 computes sur East de même paramètres, à vérifier.
Pas de variabilité sur rpact.

```
Rows: 81 Columns: 976
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr   (3): GUID, DesignID, ProductVersion
dbl (961): TestID, ToCompute__A, WhatToComp__AA, DesignType__AAA,
TrialType_ ...
lgl  (12): Rfile__AACAA, Rfunc__AACAA, Rfile__AACAB, Rfunc__AACAB,
Rfile__AA ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

Comparision